

(a)

one token stream, see (c)

Qwen3-ASR thinker LM (1.7B)decoder-only, KV cache
over the whole sessionspk.
activity
head**MLP adapter**

one embedding per 80 ms chunk

Causal FastConformer encoderNemotron 3.5, 0.6B, cache-aware
measured look-ahead ≤ 80 ms

one mono mix of both speakers

(b)

speaker A, English
(hidden reference)speaker B, German
(hidden reference)input: one audio
embedding / 80 mstranscript stream
(word end + δ_s)translation into German
(unit end + δ_t)translation into English
(unit end + δ_t)

session time (s) 0.0

unit 1

unit 2

I need a taxi to the station

turn-end overlap

unit 1 (= turn)

Welcher

Eingang?

 $+\delta_s$

A: i need a taxi to the station

Ich brauche ein Taxi

 $+\delta_t$ zum Bahnhof. *turn end: flush*

B: welcher eingang

Which entrance?

0.8

1.6

2.4

3.2

(c)

prefix (input only): ... language pair English-German $\langle \text{DELAY_2} \rangle \langle \text{TDELAY_3} \rangle$; then, per 80 ms chunk k: audio embedding, speaker-tagged transcript, target-tagged translation, $\langle \text{next} \rangle$

... a_{30} $\langle \text{A} \rangle$ station $\langle \text{next} \rangle$ a_{31} $\langle \rightarrow \text{de} \rangle$ zum Bahnhof. $\langle \text{next} \rangle$ a_{32} $\langle \text{next} \rangle$ a_{33} $\langle \text{B} \rangle$ welcher $\langle \text{next} \rangle$... a_{44} $\langle \rightarrow \text{en} \rangle$ Which entrance? $\langle \text{next} \rangle$...

a_k audio embedding (input) $\langle \text{next} \rangle$ end of chunk $\langle \text{A} \rangle$ speaker tag word transcript $\langle \rightarrow \text{de} \rangle$ target tag Taxi translation tint: speaker's language — training target dashed: planned